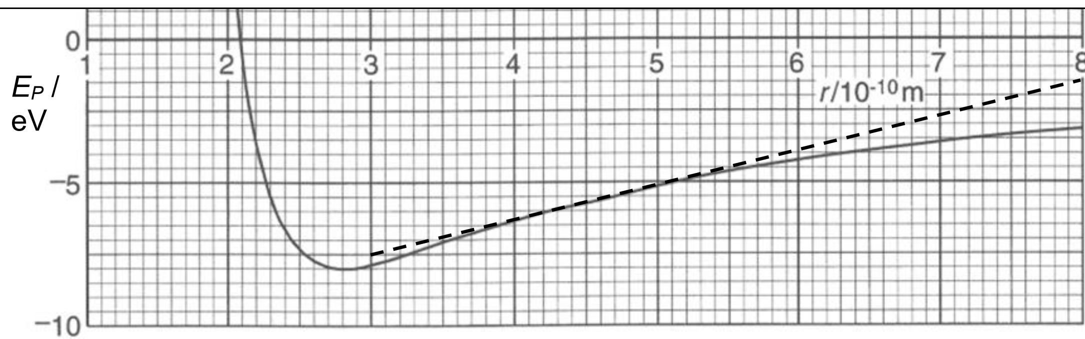
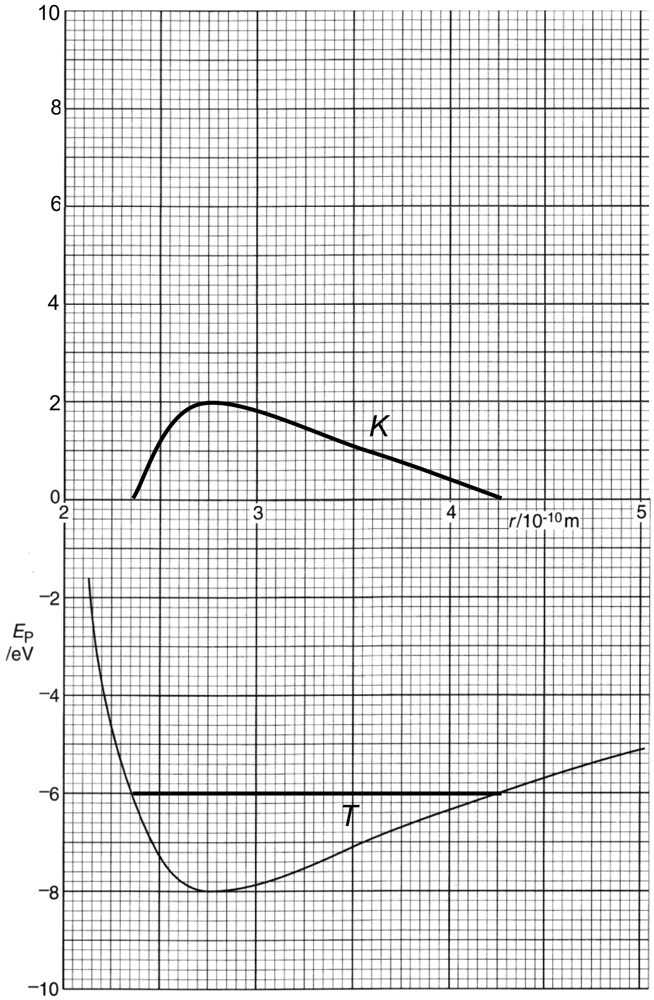
7ai	Work is the product of a force and the distance moved in the direction of the force. $W = F \times \Delta r$ Potential energy E_P is the work done (by external force in moving a point charge from infinity to the point.) *($E_P = -F \times \Delta r$)* Gradient of the graph G $= \dfrac{\Delta E_P}{\Delta r} = -F$ (accept if no negative sign) Hence the magnitude of $F = G$
7aii	For $r < 2.8 \times 10^{-10}$ m, the gradient is negative, hence the force is positive and thus repulsive force. For $r > 2.8 \times 10^{-10}$ m, the gradient is positive, hence the force is negative and thus attractive force.
7aiii1	At 2.8×10^{-10} m, gradient is zero. $F = 0$ N
7aiii2	 At 5.0×10^{-10} m, Gradient = $\dfrac{[-1.5-(-7.5)]\times 1.60\times 10^{-19}}{(8-3)\times 10^{-10}}$ $= 1.9 \times 10^{-9}$ $F = 1.9 \times 10^{-9}$ N

7b	1. The force causes <u>repulsion</u> between the ions. (This is because the gradient of the graph of the equation $E = \frac{B}{r^8}$ is negative, hence, the force $F = -\frac{dE_P}{dr}$ is positive) 2. The force <u>is significant when the ions are very close to each other</u>, and <u>decreases rapidly as the distance increases</u>. (As seen from the very steep gradient when the ions are close to each other).
7ci	 1. Horizontal line at –6 eV for total energy, labelled *T*. 2. Reasonable curve starting and ending at zero, with a maximum point Curve's maximum point is at (2.75, 2.0), since potential energy is minimum at that point. By right, K line should be drawn such that at every point, KE + PE = 6eV.
7cii	Minimum value of 2.35×10^{-10} m. Maximum value of 4.25×10^{-10} m
7ciii	*F* is not directly proportional to displacement *r*. OR the graph is not symmetrical Hence, not simple harmonic.

7d	As the lattice is heated, its total energy increases. The total energy of the ions intersects the graph Fig. 7.3 at higher levels, giving rise to larger changes in potential energy during the oscillations. The ions will vibrate between greater values of separation *r*.